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APPLIED SCIENCE

# SUPPLEMENT GUIDE

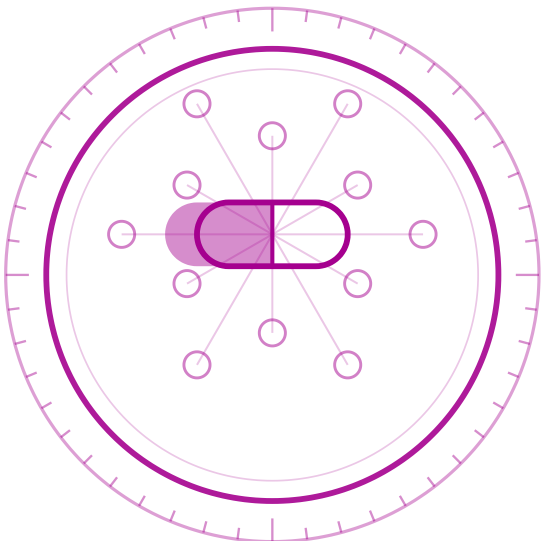
What to take, how much, when and why — and above all, **what never to buy**.

5 ESSENTIALS

4 OPTIONAL

10 THAT DO NOT WORK

REAL BUDGET



SUPPLEMENTS  
Dose and timing

|                      |                    |                       |            |
|----------------------|--------------------|-----------------------|------------|
| 9<br>EVIDENCE-BACKED | 10<br>MONEY WASTED | ~\$55<br>MONTHLY COST | 8<br>PAGES |
|----------------------|--------------------|-----------------------|------------|

## BEFORE YOU SPEND A SINGLE EURO

Supplements contribute, at most, **5 % of the result**. If the other 95 % is not in place, no tub is going to fix it. This is the real hierarchy, in order of impact.

### 01 Calorie balance and protein

They determine whether you gain or lose, and how much of it is muscle.

### 02 Training with progressive overload

The stimulus without which there is no adaptation.

### 03 7-9 hours of sleep

Where recovery and protein synthesis actually happen.

### 04 Consistency over months

The factor that most separates those who progress from those who do not.

### 05 Supplementation

The fine tuning. It only makes sense once the above already works.

## WHAT IS IN THIS GUIDE

### 02

#### THE 2 ESSENTIALS

Protein and creatine: dose, timing and brands

P. 3

### 03

#### THE 3 FOR HEALTH AND RECOVERY

Omega-3, vitamin D3 + K2 and magnesium

P. 4

### 04

#### SECOND TIER: PERFORMANCE

Caffeine, beta-alanine, citrulline and ashwagandha

P. 5

### 05

#### THE 10 THAT DO NOT WORK

Where most people's money goes

P. 6

### 06

#### YOUR DAILY PROTOCOL AND THE REAL COST

Timing table and monthly budget

P. 7

### 07

#### SAFETY, QUALITY AND ANTI-DOPING

How to choose without taking risks

P. 8

#### THE QUESTION TO ASK YOURSELF IN FRONT OF ANY TUB

Is there at least one meta-analysis backing its effect in healthy humans, at the dose the product actually contains? If the answer is no, you are paying for marketing.

## THE TWO ESSENTIALS

### 1 · PROTEIN POWDER (WHEY OR PLANT-BASED)

It is not magic: 1 g of whey protein builds exactly the same as 1 g of protein from chicken. Its value is **logistical** — it lets you hit your daily quota when real food does not fit your schedule.

| PARAMETER               | DETAIL                                                              |
|-------------------------|---------------------------------------------------------------------|
| <b>Dose</b>             | 20–40 g per serving (1–2 scoops)                                    |
| <b>When</b>             | Post-workout or between meals, as a top-up to close the daily total |
| <b>Type</b>             | Concentrate (cheaper) or isolate (less lactose and fat)             |
| <b>Vegan option</b>     | Pea + rice in a 70/30 ratio, to complete the amino acid profile     |
| <b>Verified brands</b>  | Optimum Nutrition, MyProtein, Garden of Life (vegan)                |
| <b>Indicative price</b> | \$30–60 for 2 kg (1 to 2 months of use)                             |

### 2 · CREATINE MONOHYDRATE

The most studied supplement in the history of sport, with hundreds of trials and an excellent safety profile in healthy people. It raises strength by 5 to 15 %, speeds recovery between sets and has documented cognitive benefits.

| PARAMETER               | DETAIL                                                                           |
|-------------------------|----------------------------------------------------------------------------------|
| <b>Dose</b>             | 3–5 g a day. The loading phase is optional and does not improve the final result |
| <b>When</b>             | Any time of day. Taking it with a meal slightly improves absorption              |
| <b>Type</b>             | <b>Monohydrate only.</b> The other forms are more expensive marketing            |
| <b>Cycling</b>          | Not needed. Continuous use is safe and more effective                            |
| <b>Brands</b>           | Creapure (Germany) or any unflavoured monohydrate from a known brand             |
| <b>Indicative price</b> | \$15–25 for 500 g (3–4 months of use)                                            |

#### TWO CLARIFICATIONS ABOUT CREATINE

The 1–2 kg gain of the first weeks is **intramuscular water**, not fat: it is the intended effect. And no, it does not damage the kidneys in healthy people — it can slightly raise creatinine in a blood test, which is an artefact of the measurement, not harm. Tell your doctor you take it before any blood work.

## THE THREE FOR HEALTH AND RECOVERY

### 3 · OMEGA-3 (EPA + DHA)

| PARAMETER               | DETAIL                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------|
| <b>Dose</b>             | 2–3 g of EPA + DHA combined per day (check the back: that is not the same as grams of oil) |
| <b>Benefits</b>         | It lowers inflammation, improves recovery and protects cardiovascular health               |
| <b>When</b>             | With meals containing fat, to maximise absorption                                          |
| <b>Type</b>             | Fish oil (salmon, sardine, anchovy) or algae oil for the vegan option                      |
| <b>Brands</b>           | Nordic Naturals, Solgar, OmegaVia                                                          |
| <b>Indicative price</b> | \$20–35 for 90 capsules                                                                    |

### 4 · VITAMIN D3 + K2

A large share of the population has insufficient levels, especially with indoor work or in cloudy climates. It is involved in immune function, bone health, recovery and hormone production.

| PARAMETER               | DETAIL                                                                        |
|-------------------------|-------------------------------------------------------------------------------|
| <b>D3 dose</b>          | 2,000–4,000 IU daily                                                          |
| <b>K2 dose</b>          | 100–200 mcg (it directs calcium to bone rather than to the artery)            |
| <b>When</b>             | In the morning, with breakfast: it is fat-soluble and needs fat               |
| <b>Form</b>             | D3 (cholecalciferol), never D2 — it is absorbed far worse                     |
| <b>Monitoring</b>       | Ask for 25-OH-vitamin D in your annual blood test and adjust with your doctor |
| <b>Indicative price</b> | \$10–20 for 3 months                                                          |

### 5 · MAGNESIUM

| PARAMETER               | DETAIL                                                                             |
|-------------------------|------------------------------------------------------------------------------------|
| <b>Dose</b>             | 300–400 mg of elemental magnesium a day                                            |
| <b>Benefits</b>         | It improves sleep quality and neuromuscular function, and helps modulate cortisol  |
| <b>When</b>             | 30–60 minutes before sleep                                                         |
| <b>Form</b>             | Bisglycinate (better tolerated) or citrate. Avoid the oxide: it is barely absorbed |
| <b>Indicative price</b> | \$15–25 for 3 months                                                               |

#### BUYING ORDER IF YOUR BUDGET IS LIMITED

Creatine → protein → vitamin D3 → magnesium → omega-3. The first two have the greatest impact per euro spent in an athlete who already trains and eats well.

## SECOND TIER: PERFORMANCE

They have solid evidence, but their effect is smaller and more specific. Add them only once the previous five are already part of your routine.

| SUPPLEMENT                | DOSE                    | WHEN                  | WHAT TO REALISTICALLY EXPECT                                        |
|---------------------------|-------------------------|-----------------------|---------------------------------------------------------------------|
| <b>Caffeine</b>           | 3–6 mg/kg               | 45–60 min before      | More strength, more reps and a lower perception of effort           |
| <b>Beta-alanine</b>       | 3–5 g/day               | Any time, daily       | It improves efforts of 1 to 4 minutes. It needs 4 weeks to saturate |
| <b>Citrulline malate</b>  | 6–8 g                   | 60 min before         | A few more reps and a better muscle pump                            |
| <b>Ashwagandha KSM-66</b> | 300–600 mg              | Night                 | It helps with perceived stress. Moderate evidence, not spectacular  |
| <b>Electrolytes</b>       | According to sweat rate | Long sessions or heat | It prevents cramps and slumps when you sweat a lot                  |

### CAFFEINE: HOW TO USE IT WELL

#### WHAT TO DO

- Start at 3 mg/kg and go up only if you need to
- Combine it with L-theanine in a 1:2 ratio if it makes you jittery
- Save it for your 2–3 hardest sessions of the week
- Take a week off every 6–8 to restore sensitivity

#### WHAT RUINS THE RESULT

- Taking it after 14:00: its half-life is 5–6 hours
- Using it daily at high doses: tolerance cancels the effect
- Using caffeine to paper over insufficient sleep
- Stacking pre-workout, coffee and an energy drink without counting the total

#### SAFETY LIMIT

Do not exceed 400 mg of caffeine a day (about 4 coffees), nor 200 mg in a single dose if you have no tolerance. If you have hypertension, arrhythmias, anxiety or sleep problems, check with your doctor before using it as an ergogenic aid.

A note on beta-alanine: the tingling in face and hands (paraesthesia) is harmless and goes away by splitting the dose in two or using the sustained-release version.

## THE 10 THAT DO NOT WORK

None of these products is dangerous in itself; they simply do not do what the label promises. This is the money most people throw away every month.

| PRODUCT                                   | WHAT IT PROMISES                   | WHAT THE EVIDENCE SAYS                                                                 |
|-------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------|
| <b>BCAAs / branched-chain amino acids</b> | More muscle and less catabolism    | Useless if you already get enough protein: the other essential amino acids are missing |
| <b>Thermogenic fat burners</b>            | Speeding up metabolism             | The effect on expenditure is minimal and does not change body composition              |
| <b>Glutamine</b>                          | Recovery and hypertrophy           | No effect in healthy, well-fed athletes                                                |
| <b>Testosterone boosters</b>              | Raising testosterone               | Tribulus and the like do not raise it in men with normal levels                        |
| <b>«Advanced» creatines</b>               | Better absorption than monohydrate | HCL, ethyl ester and kre-alkalyn: more expensive, with no proven advantage             |
| <b>Detoxes and cleanses</b>               | Removing toxins                    | Your liver and kidneys already handle that, free of charge                             |
| <b>Nitric oxide capsules</b>              | Pump and vasodilation              | They usually contain doses far below the effective amount of citrulline                |
| <b>Megadose multivitamins</b>             | Covering every deficiency          | On a varied diet you only produce expensive urine; some vitamins accumulate            |
| <b>Collagen for muscle</b>                | Gaining muscle mass                | A low biological value protein. Useful for tendon, not for hypertrophy                 |
| <b>Exogenous ketones</b>                  | Burning fat and giving energy      | Expensive, unpleasant tasting and of no benefit to an athlete who eats carbohydrates   |

### THE 30-DAY RULE

Before buying a new supplement, work out what it costs per year and ask yourself what else you would do with that money: a few more months of creatine and protein? Better-quality real food? Almost always the right answer is the second.

YOUR DAILY PROTOCOL

| SUPPLEMENT              | MORNING        | PRE-WORKOUT      | POST-WORKOUT     | NIGHT                 |
|-------------------------|----------------|------------------|------------------|-----------------------|
| Protein powder          | —              | —                | CORE             | If the quota is short |
| Creatine 3-5 g          | OPTION         | Option           | WITH THE PROTEIN | Option                |
| Omega-3                 | WITH BREAKFAST | —                | —                | Option                |
| Vitamin D3 + K2         | REQUIRED       | —                | —                | —                     |
| Magnesium               | —              | —                | —                | 30 MIN BEFORE SLEEP   |
| Caffeine (optional)     | —              | 45-60 MIN BEFORE | —                | Never                 |
| Beta-alanine (optional) | Option         | Option           | Option           | Option                |

A REALISTIC MONTHLY BUDGET

| LEVEL        | INCLUDES                             | MONTHLY COST |
|--------------|--------------------------------------|--------------|
| Bare minimum | Creatine + vitamin D3                | ≈ \$10       |
| Recommended  | Creatine + protein + D3 + magnesium  | ≈ \$40       |
| Full         | The 5 essentials + caffeine          | ≈ \$55       |
| Performance  | Complete + beta-alanine + citrulline | ≈ \$75       |

YOUR INVENTORY

| SUPPLEMENT      | BRAND | DAILY DOSE | RESTOCK DATE |
|-----------------|-------|------------|--------------|
| Protein         |       |            |              |
| Creatine        |       |            |              |
| Omega-3         |       |            |              |
| Vitamin D3 + K2 |       |            |              |
| Magnesium       |       |            |              |

## WHERE EVERY NUMBER COMES FROM

Every number in this document comes from sources that are public, free and checkable. Not one figure is invented, rounded for convenience or copied from another popular publication. This page exists so that you can verify it yourself.

### WHY THESE SOURCES AND NO OTHERS

Works of the United States federal government are not subject to copyright (17 U.S.C. § 105). That makes a handful of agencies the only source of nutrition and training data that is both first-rate and legally free. This is not restricted material: it is public material that almost nobody opens.

| WHAT YOU ARE LOOKING FOR     | WHAT YOU WILL FIND                                                           | WHERE                        |
|------------------------------|------------------------------------------------------------------------------|------------------------------|
| Reference intakes            | RDA, AI and tolerable upper intake level for every vitamin and mineral       | ods.od.nih.gov               |
| Food composition             | The full profile of any food, nutrient by nutrient                           | fdc.nal.usda.gov             |
| Dietary guidelines           | Dietary patterns and limits for added sugars, saturated fat and sodium       | health.gov                   |
| Physical activity guidelines | The weekly dose, its response curve and the strength requirement             | health.gov                   |
| Physical readiness doctrine  | FM 7-22: training, nutrition, sleep and mental readiness                     | armypubs.army.mil            |
| Lifting equation             | How much weight is reasonable to lift in each specific situation             | cdc.gov/niosh                |
| Food safety                  | Safe internal temperatures and the danger zone                               | foodsafety.gov               |
| Health claims                | What may legally be claimed about a nutrient in Europe, and what was refused | European Commission register |

### HOW THIS DOCUMENT CHECKS ITSELF

The reference figures are not typed by hand: they live in a data module that the typesetting program verifies before writing a single line. It checks that the macronutrient ranges can add up to one hundred per cent of energy, that none of the 19 micronutrients has a recommended value above its own ceiling without a note explaining why, and that no safe cooking temperature falls inside the danger zone. If a single check fails, the document is not typeset.

### REFERENCE VALUES ARE POPULATION VALUES

They describe what covers the needs of most healthy adults. They are not an individual prescription, they do not account for disease, pregnancy, breastfeeding or medication, and no publication replaces assessment by a registered health professional. If you take medication, some of the nutrients named here interact with it in clinically relevant ways: do not change an intake or start a supplement without telling whoever prescribes for you.



## QUALITY, SAFETY AND ANTI-DOPING

### HOW TO CHOOSE A RELIABLE PRODUCT

- Third-party testing marks: **Informed Sport** or **NSF Certified for Sport**
- A label with the dose per serving, not a «proprietary blend» with no amounts
- A single active ingredient per tub whenever possible
- A manufacturer with batch, expiry date and country of origin visible

### WARNING SIGNS

- Promises of results «in 7 days» or comparisons with drugs
- Proprietary formulas that hide the real doses
- Weight-loss products bought outside regulated channels
- Ingredients you can neither pronounce nor find in the literature

### SITUATIONS THAT REQUIRE CHECKING WITH A HEALTH PROFESSIONAL FIRST

- ☐ You take long-term medication: anticoagulants, antihypertensives, antidiabetics, thyroid drugs or psychotropics
- ☐ You have renal, hepatic or cardiac disease, or an eating disorder
- ☐ You are pregnant, breastfeeding or a minor
- ☐ You compete under anti-doping control: check every product against the WADA list
- ☐ You are about to have blood work: mention that you take creatine

### FOOD FIRST, THE TUB SECOND

No supplement in this guide makes up for sleeping badly or eating too little: **they are the final 5 %**, not the shortcut. Start with the two essentials, give them eight weeks, and add nothing else until you can measure what changed.

#### BASIC

**\$9.99/month**

Fundamentals  
and quick guides

#### PRO

**\$19.99/month**

Your current plan  
Advanced nutrition

#### ELITE

**\$34.99/month**

Everything included  
+ coaching workbook

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#### IMPORTANT NOTICE

This guide is for informational and educational purposes only. It does not constitute medical advice, diagnosis, treatment or prescription, and it does not replace assessment by a registered health professional. Food supplements are not intended to prevent, treat or cure any disease, and must not replace a varied, balanced diet. The doses given are for healthy adults; consult your doctor or pharmacist before starting any supplementation, especially if you take medication, have a medical condition, are pregnant or breastfeeding, or are a minor. The brands named are mentioned for guidance only and imply no sponsorship or commercial relationship. Prices are indicative and may vary by country and retailer. Copyrighted content: reproduction, distribution or resale without written permission from VILLUMINATIONS is prohibited.